

Measuring Validator and Sandboxed-Execution Coverage for LLM-Generated NetOps Artifacts: A Paired Confirmatory Study

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Abstract—We preregistered and executed a paired confirmatory experiment (study_id f2c3d8b1-7a6e-4a9a-b2c3-5f8e1d2b6c7e) that measured the marginal detection benefit of adding sandboxed emulation to a deterministic validator for LLM-generated intent→configuration artifacts. Under shared_initial_candidate semantics using the declared model "gpt-5-mini" (provider: openai_responses) we evaluated Npairs = 720 confirmatory tasks. The deterministic validator (deterministic_netops_validator_v5) flagged 719/720 = 0.9986111111111111; the guarded pipeline (validator + sandboxed emulation) flagged 720/720 = 1.0. Paired difference (guarded - baseline) = 0.001388888888888884; exact McNemar two-sided p = 1.0; 95% bootstrap percentile CI (10000 resamples, seed = 1729) = [0.0, 0.004166666666666667]. Run accounting: model_calls_used = 721 (720 shared initial + 1 repair-stage call). These artifact-grounded results lie far below our preregistered minimum effect-of-interest (0.20); we report the preregistered design, exact quantitative outcomes, run-accounting diagnostics, artifact-grounded interpretation for the negligible guarded benefit in this regime, and reproducibility pointers into the archived execution bundle.

I. INTRODUCTION

Generative large language models (LLMs) are increasingly proposed to translate operator intent into device configuration and NetOps workflows. Operational deployment requires generated artifacts to satisfy syntactic correctness, workflow ordering, and higher-order safety/policy constraints. A common mitigation architecture is generate→validate→(repair)→execute, sometimes augmented by sandboxed emulation to reveal runtime-only failures that static validators miss. Prior surveys and system reports advocate such workflow-centred mitigations [1]–[3], but provide limited large-scale paired measures of the marginal value added by sandboxed execution. Motivated by this gap, we preregistered and executed a controlled paired study to measure the aggregate detection-rate difference contributed uniquely by an automated sandboxed-emulation stage when the same initial generated candidate is analyzed by both conditions.

Research question and contributions. Our preregistered research question was: for synthetic intent→configuration tasks under shared_initial_candidate semantics, what is the paired detection-rate difference (guarded - baseline) between (A) a deterministic validator-only pipeline and (B) the same deterministic validator followed by automated sandboxed emulation? Contributions: (1) a machine-readable preregistration and faithful autonomous execution; (2) an artifact-grounded

paired analysis on Npairs = 720 with complete accounting; (3) an artifact-grounded interpretation explaining a near-zero marginal effect; (4) concise reproducibility pointers into the archived bundle and prescriptive boundary conditions where guarded stages are likely to matter.

II. RELATED WORK

Workflow-centred mitigations and verification tools. Surveys and recent system reports document architectures that pair LLM generation with validators, verifiers, and emulators to improve correctness of generated NetOps artifacts and adjacent program-generation tasks [1], [2]. Tool-focused work demonstrates generate→validate→repair patterns (e.g., Batfish/NetKAT style verification, emulator-assisted validation) but reports heterogeneous gains depending on task, validator rule coverage, and emulator fidelity [2], [3].

Empirical findings motivating a paired design. Empirical studies in program/specification generation show external verifiers often catch model mistakes not found by the model alone, and some NetOps prototypes use emulation to expose runtime issues [3]. These prior findings motivated our preregistered paired design that isolates guarded-stage contributions: because we enforce shared_initial_candidate semantics, the only permissible sources of between-condition discordance are downstream guarded-stage emulation observations and permitted repair-stage invocations recorded in run accounting.

III. METHODOLOGY

Preregistration and experimental design. We preregistered a confirmatory paired-binary design (study_id f2c3d8b1-7a6e-4a9a-b2c3-5f8e1d2b6c7e). Primary estimand: paired_success_rate_difference_guarded_minus_baseline aggregated across confirmatory samples. Planned confirmatory sample: Npairs = 720 (planned episodes = 1440). Minimum effect-of-interest for power planning = 0.20 (20 percentage points). The preregistration specified inclusion/exclusion rules, missingness handling, multiplicity handling for exploratory secondary tests, and deterministic contamination controls.

Task bank, vendors, and inclusion. Confirmatory items were synthetic intent→configuration tasks from intent_configuration_repair_v1 covering three abstracted vendor-CLI families and three topology templates (leaf-spine, 3-node service chain, triangle). Inclusion required vendor-specific parser acceptance; the

preregistered missingness policy permitted up to two deterministic reseeds for parser failures; irrecoverable parser failures would be deterministically excluded and replaced. In execution there were zero irrecoverable parser exclusions: `complete_pair_count = 720` (see `analysis/missingness_summary.csv`).

Model configuration and sampling semantics. Declared/requested model: "gpt-5-mini" (provider field: `openai_responses`). The archived model configuration records `max_output_tokens = 2000` and `temperature = null` (`execution/model_configuration.json`, sha256 a40e96e212ab87da...). Important clarification (preregistered and enforced): `temperature = null` indicates the client did not set an explicit temperature parameter; `null` does not imply `temperature=0` or provider-side determinism. We enforced `shared_initial_candidate` semantics: for each pair we obtained exactly one sampled initial candidate (shared) which both baseline and guarded pipelines analyzed. Therefore any between-condition difference can only arise from (i) guarded-stage emulation observations not encoded as a validator rule, or (ii) the allowed downstream repair-stage invocation(s) recorded in run accounting.

Validator and guarded pipeline implementation. Baseline: `deterministic_netops_validator_v5` performing full intent-constraint validation (syntax, command ordering/workflow, required-field presence, and policy-level constraints) and returning binary detected/not-detected per the preregistered scoring rule `full_intent_constraint_validation`. Guarded: baseline validator followed by an automated sandboxed-emulation harness executing the candidate against an emulated instance of the declared topology and producing runtime observations. The repair policy permitted at most one repair-stage model call per task; run accounting records exactly one repair-stage call in the confirmatory execution.

Execution controls, provenance, and deterministic reconciliation. The adapter family used: `hosted_netops_gvr_v1`. Execution manifest and deterministic reconciliation artifacts record `completed_episode_count = 1440`, `complete_pair_count = 720`, `failed_episode_count = 0`, and provider-call audit information including `model_calls_used = 721` and `stage_counts: shared_initial_generation = 720; repair = 1`. Provenance and contamination controls (deterministic hashing, artifact-version checks) were applied per the preregistration; no contamination flags occurred for the confirmatory set.

IV. RESULTS

Primary confirmatory counts and test statistics (artifact-grounded). Analysis artifacts report the paired contingency counts: `n11 = 719` (detected by both), `n10 = 1` (guarded-only), `n01 = 0` (baseline-only), `n00 = 0` (neither). Marginal detection rates are `baseline = 719/720 = 0.9986111111111111` and `guarded = 720/720 = 1.0`. The paired estimate (`guarded - baseline`) = `0.00138888888888888884`. Exact McNemar two-sided test `p = 1.0`. The paired-bootstrap percentile 95% CI (10000 resamples, `bootstrap_seed = 1729`) = `[0.0, 0.004166666666666667]`.

These numbers are reproduced verbatim from `analysis/condition_summary.csv` (sha256 394bc690671b731194f9b42b4dcbe28fdd6e2ec8cf898b79831225343c22c28) and `analysis/paired_contingency_table.csv` (sha256 efe6e8e7016fa843a8226e911dbde829e943903d719101da8d956b3ff899133c).

Canonical artifact pointers and the unique discordant episode. The complete per-episode raw results are archived in `execution/raw_results.jsonl` (input-results SHA-256 = `df367ecc40a6f0eaf402129260e11d8450884a4ea5fdb94a602e0d2513aef0e8`). The single discordant pair (the unique `n10` item) is the lone entry in `execution/raw_results.jsonl` whose recorded `baseline_score == 0` and `guarded_score == 1`. That `raw_results` entry contains the canonical `pair_id` field and includes the `validator_trace` and `repair_provider_trace` fields that link the guarded detection to a downstream repair-stage provider trace. Deterministic reconciliation and provider-call accounting are archived in `analysis/deterministic_reconciliation.json` (sha256 49731a2e52e049ccf5b53c35ddac2351ad75e7e95504dc4a42bcb096d0dec3aa) and confirm `model_calls_used = 721` and `stage_counts = {"shared_initial_generation": 720, "repair": 1}`. In short: the discordant episode is uniquely identifiable within the archived bundle (`execution/raw_results.jsonl`, sha256 `df367ecc...`) and the same bundle contains the `deterministic_reconciliation` artifact (sha256 49731a2e5...) that documents the single repair-stage call that is connected to that `raw_results` entry via its `repair_provider_trace` field.

Run-accounting diagnostics and sensitivity checks. Provider-call audit (`deterministic_reconciliation.json`) reports `hosted_model_call_event_count = 721`, `completed_hosted_model_call_event_count = 721`, `failed_hosted_model_call_event_count = 0`, `cache_hit_count = 0`, `cache_miss_count = 721`. The preregistered missingness policy and sensitivity analyses were applied; no irrecoverable parser or emulation failures occurred (`analysis/missingness_summary.csv` documents `complete_pairs = 720` and zeros for all failure categories). The primary analysis used the preregistered `complete_pair_primary` treatment; the preplanned sensitivity (treating irrecoverable technical failures as nondetections) is moot here because there were no such failures in the confirmatory run.

Representative per-episode diagnostics. Representative scoring traces (`execution/scoring/task-000001-*.json ... task-000006-*.json`) show `parsed_command_count` values of 4, 2, 6, 4, 3, and 6 respectively and `validator_violation_count = 0` for these examples; the `validator_version` reported in these traces is "5.0". These representative artifacts illustrate that the `shared_initial` candidates in the confirmatory bank were typically complete with respect to the required command sequences and that validator checks reported validity for the majority of cases. The representative traces are archived in the execution bundle and are consistent with the high baseline success rate.

Attribution of the single discordant detection. Because `shared_initial_candidate` semantics were enforced (one shared

sampled initial candidate per pair), the only proximate causes for the single n10 observation are (A) an emulation observation not encoded as a validator rule that produced a guarded-stage detection, or (B) the single recorded repair-stage invocation documented in provider-call accounting. The `raw_results` entry for the discordant pair includes the `repair_provider_trace` and `validator_trace` fields permitting an auditor to follow the exact artifact-level linkage from the guarded detection to the recorded repair-stage call; both the `raw_results` file (`sha256 df367ecc...`) and the reconciliation artifact (`sha256 49731a2e5...`) are present in the canonical execution bundle.

Implications of the quantitative diagnostics. The observed discordant count ($n_{10} = 1$) yields an aggregate paired difference that is two orders of magnitude below our preregistered minimum effect-of-interest (0.20). The bootstrap CI [0.0, 0.004166666666666667] quantifies the narrowness of the realized uncertainty in this curated regime. Combined with run-accounting (`model_calls_used` = 721; `repair` = 1) and the representative per-episode diagnostics, the archived evidence supports a ceiling-effect interpretation: the deterministic validator covered the task bank’s explicit syntactic and workflow constraints so thoroughly that the guarded-stage emulation had almost no additional aggregate detections to surface.

V. DISCUSSION

Why was the guarded-stage aggregate benefit negligible? The archived evidence supports three artifact-grounded factors.

1) Validator ceiling effect. Deterministic `netops_validator_v5` covered the syntactic and explicit semantic properties encoded in the synthetic task bank (shutdown—configure—restore, make-before-break uplink switchover, paired MTU changes). Baseline success rate ≈ 0.9986 indicates near-universal validator coverage for these curated patterns, leaving negligible headroom for the guarded stage.

2) Task curation and inclusion policy. Confirmatory items were parser-validated synthetic tasks specifically designed to exercise workflow policies but not constructed adversarially to evade validator checks. This curation concentrates cases inside the validator’s coverage envelope and reduces the prevalence of subtle runtime-only failure modes that emulation might reveal.

3) Shared `initial_candidate` semantics and limited repair activity. Enforcing one sampled initial candidate per pair isolates guarded-stage contributions but removes between-condition generation variability. `Provider_call_audit` and `deterministic_reconciliation.json` record one repair-stage invocation aligned with the single n10 discordant count; thus the observed discordance is attributable to downstream guarded-stage activity recorded in run accounting rather than generation differences.

Operational implications and boundary conditions. For environments where a high-coverage static validator exists and inputs match its expected patterns (parser-validated, constrained intents), adding automated sandboxed emulation may provide negligible aggregate additional detection at scale.

This conditional conclusion must not be generalized: in operational regimes with noisier natural-language intents, adversarial inputs, broader vendor diversity, stateful multi-step interactions, or validator gaps, guarded runtime checks and higher-fidelity emulation are likely to provide substantive benefit. The archived per-episode traces support targeted follow-up hypotheses: adversarial task injection, broader vendor-CLI coverage, and complementary validator families should increase the guarded-stage marginal contribution.

VI. LIMITATIONS

- Synthetic task bank and deterministic task generator produce limited ecological diversity and create a validator-ceiling effect; results may not generalize to noisier, adversarial, or production distributions.
- Single declared/requested model family (`gpt-5-mini`) and single validator implementation (`deterministic_netops_validator_v5`) limit generalizability across models and validator families.
- Preregistered shared `initial_candidate` semantics (one initial generation reused across paired conditions) isolate guarded-stage contribution but remove between-condition generation variability; this design choice constrains discordance sources to downstream guarded-stage observations or repair calls.
- Sandboxed-emulation fidelity relative to vendor/OS production behaviors and large-scale stateful interactions is limited by the emulation harness used; higher-fidelity emulators could change outcomes in other regimes.
- Study power planning targeted a minimum detectable paired difference of 0.20; the observed aggregate effect (~ 0.0014) lies far below that design sensitivity and should be interpreted as a narrowly scoped near-zero finding for this experimental regime rather than a general negative result for all NetOps workflows.
- Provider-side nondeterminism and hidden caching remain residual uncertainties despite deterministic reconciliation procedures; these are documented in `analysis/deterministic_reconciliation.json` and `provider_call_audit` artifacts.

VII. CONCLUSION

We preregistered and executed a paired confirmatory study comparing a strong deterministic validator with the same validator augmented by sandboxed emulation on a curated synthetic intent→configuration task bank. Over 720 paired tasks the guarded pipeline produced a negligible aggregate detection gain (0.0013888888888888884) with exact McNemar $p = 1.0$ and 95% bootstrap CI [0.0, 0.004166666666666667]. Run accounting shows one repair-stage invocation corresponding to the single guarded-only detection; because shared `initial_candidate` semantics were enforced, archived per-episode traces permit independent verification of that mapping via `execution/raw_results.jsonl` and `analysis/deterministic_reconciliation.json`. We interpret the result as arising from validator ceiling effects and curated

task design; follow-up work should focus on adversarial and production-like task banks, multi-validator ensembles, and higher-fidelity emulation to identify regimes where guarded stages add substantive operational value.

Reproducibility pointers (compact). The canonical execution bundle archived by the run contains: execution/execution_manifest.json (execution summary and preregistration SHA-256), execution/raw_results.jsonl (input-results SHA-256 df367ecc40a6f0eaf402129260e11d8450884a4ea5fdb94a602e0d2513aef0e8), analysis/deterministic_reconciliation.json (sha256 49731a2e52e049ccf5b53c35ddac2351ad75e7e95504dc4a42bcb096d0dec3aa), execution/model_configuration.json (sha256 a40e96e212ab87da251ac0d6dbb75bc543afa13438b5a6b9ebd073597ffdd820), analysis/paired_contingency_table.csv (sha256 efe6e8e7016fa843a8226e911dbde829e943903d719101da8d956b3ff899133c), and analysis/condition_summary.csv (sha256 394bc690671b731194f9b42b4dcbe28fded6e2ec8cf898b79831225343c22c28).

The discordant episode is the unique raw_results.jsonl entry with baseline_score = 0 and guarded_score = 1; that record contains the pair_id and validator/emulation traces that map it to the single repair-stage invocation recorded in deterministic_reconciliation.json. These artifacts are archived as the canonical reproducibility bundle referenced by the execution manifest and the preregistration record. (See Disclosure for exact manifest references.)

DISCLOSURE STATEMENT

Disclosure: Autonomous post-lock research run executed under the frozen master prompt supplied to the system. Declared/requested model: "gpt-5-mini" (provider recorded as openai_responses). Deterministic validator: deterministic_netops_validator_v5. Orchestration adapter family: hosted_netops_gvr_v1. Preregistration id: f2c3d8b1-7a6e-4a9a-b2c3-5f8e1d2b6c7e (preregistration SHA-256: 0169919fb8c5aedb2c5a78e031f1ab5c3aada405cde1ae80fb2f864768595c1b; recorded in the execution manifest). Initial master-prompt immutable reference: provenance/master_prompt.txt, SHA-256 = 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15ba. Execution summary (artifact-grounded): completed_episode_count = 1440, complete_pair_count = 720, model_calls_used = 721 (720 shared initial + 1 repair-stage call); stage_counts and provider-call audit are archived in analysis/deterministic_reconciliation.json (sha256 49731a2e52e0...). Human role and autonomy boundary: a human Research Director selected the broad NetOps topic family and constrained the pre-lock master prompt; after the autonomy lock the agentic pipeline autonomously performed literature selection, preregistration, code and prompt generation, experiment execution, analysis, peer review, and manuscript drafting. No manual human editing of technical manuscript content occurred after the autonomy lock. The execution manifest and archived artifacts named above serve as the canonical reproducibility bundle.

REFERENCES

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- [2] Matt Brown, Ari Fogel, Daniel Halperin, Victor Heorhiadi, Ratul Mahajan, Todd Millstein, "Lessons from the evolution of the Batfish configuration analysis tool", 2023, 10.1145/3603269.3604866
- [3] <http://arxiv.org/abs/2407.08249>